

Working with Fractions

CHAPTER

08

Learning Outcomes...

- Fractions
- Multiplication of Fractions
- Multiplication of Fractions using Area Model
- Multiplication of Fractions-Simplifying to Lowest Form
- A Glimpse into Ancient Indian Mathematics
- Comparing Products in Fraction Multiplication
- Order of Multiplication
- Reciprocal of a Fraction
- Division of Fractions
- Comparing Quotient in Fraction Division
- Fractional Relations
- Fractions in Ancient India

Introduction

In the previous class, we have learned that fractions represent equal parts of a whole. We studied fractional units, how to read and locate fractions on a number line, and how to find equivalent fractions and simplify them to their lowest terms. We also explored Brahmagupta's methods for adding and subtracting fractions using common denominators.

In this chapter, we will extend our understanding by learning how to multiply and divide fractions, and discover how these rules were explained by ancient Indian mathematicians like Brahmagupta and Bhāskara II.

Fractions

The number of the form a/b , where $b \neq 0$ and a, b are whole numbers is called a fraction. For fraction a/b , a is called numerator and b is called denominator of the fraction.

Multiplication of Fractions

Multiplication of a Fraction by a Whole Number

To multiply a fraction by a whole number, multiply the numerator of the fraction with that whole number keeping the denominator same.

For example ; $5 \times \frac{1}{4} = \frac{5 \times 1}{4} = \frac{5}{4}$ and $7 \times \frac{2}{3} = \frac{7 \times 2}{3} = \frac{14}{3}$

Multiplication of a Fraction by a Fraction

To multiply a fraction by a fraction, multiply their numerators and denominators.

Product of two fractions = $\frac{\text{Product of numerators}}{\text{Product of denominators}}$

If $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then $\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$

For example ; $\frac{1}{2} \times \frac{4}{7} = \frac{1 \times 4}{2 \times 7} = \frac{4}{14}$ and $\frac{5}{9} \times \frac{1}{3} = \frac{5 \times 1}{9 \times 3} = \frac{5}{27}$

This method, first clearly stated by the Indian mathematician Brahmagupta in 628 CE, makes multiplying fractions simple and systematic.

Note :

- (i) To multiply mixed fraction with a fraction, first convert mixed fraction into improper fraction.
- (ii) Fraction as an operator "of" represents multiplication.

ILLUSTRATIONS

1 Find $\frac{1}{3}$ of 4 and show diagrammatically.

Soln.: $\frac{1}{3}$ of 4 = $\frac{1}{3} \times 4 = \frac{4}{3} = 1\frac{1}{3}$



2 In a class of 60 students, $\frac{1}{5}$ of the total number of students like to study English, $\frac{2}{5}$ of the total like to study Science and the remaining like to study Mathematics.

- (i) How many students like to study Mathematics, Science and English ?
- (ii) What fraction of the total students like to study Mathematics?

Soln.: Total number of students = 60

(i) Number of students who like to study English = $\frac{1}{5} \times 60 = 12$

Number of students who like to study Science = $\frac{2}{5} \times 60 = 24$

Number of students who like to study Mathematics = $60 - (12 + 24) = 60 - 36 = 24$

(ii) Required fraction = $\frac{24}{60}$

Multiplication of Fractions using Area Model

When we multiply two fractions, we are finding a part of a part. The area model helps us understand this idea using a simple rectangle. Let's understand with an example.

Let us find the product $\frac{1}{2} \times \frac{1}{3}$ by using rectangle.

Step 1: Draw a rectangle to represent 1 whole unit.



Step 2: Now divide the rectangle vertically into 3 equal parts. Shade one of those three parts to represent $\frac{1}{3}$.

So far, the shaded part = $\frac{1}{3}$ of the whole.



Step 3: Next, divide the rectangle horizontally into 2 equal parts. Now, shade half of the already shaded $\left(\frac{1}{3}\right)$ part in a different colour or pattern.



The overlapping region (where both shadings meet) shows $\frac{1}{2}$ of $\frac{1}{3}$.

Step 4: Now, the rectangle is divided into 6 small equal parts. Only 1 of those 6 parts is double-shaded.

That means the product $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$ i.e., Area = Length $\times$ Breadth

So, the area of the shaded part (the overlap) represents the product of the two fractions.

Thus, the multiplication of fractions tells us how much of the whole is represented when we take one fraction of another.

ILLUSTRATIONS

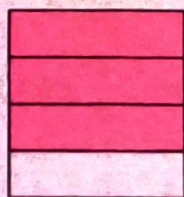
3 Find $\frac{2}{3} \times \frac{3}{4}$ using the area model.

Soln.: Step 1: Draw a rectangle to represent 1 whole unit.

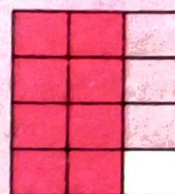


Step 2: Divide the rectangle into 4 equal horizontal strips and shade 3 out of 4 parts.

The shaded region represents $\frac{3}{4}$ of the whole.



Step 3: Now, divide the rectangle into 3 equal vertical columns. Shade 2 out of 3 columns in a different pattern. This represents $\frac{2}{3}$ of the whole.



Step 4: The part that is shaded both ways (the overlapping area) represents $\frac{2}{3}$ of $\frac{3}{4}$.

Now, the number of small rectangles in the whole figure = $3 \times 4 = 12$

The overlapping area covers $2 \times 3 = 6$ parts

So, $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$

Multiplication of Fractions-Simplifying to Lowest Form

When we multiply fractions, we multiply the numerators together and the denominators together. But before multiplying, we can make multiplication easier by simplifying the fraction that means dividing the numerator and denominator by their common factors to reduce the fraction to its simplest form.

For example ; To multiply fraction $\frac{4}{20}$ with any other fraction, we first divide the numerator and denominator by common factor 4 so that $\frac{4}{20} = \frac{1}{5}$

Also, sometimes two other fractions have common factor.

For example; For the fractions $\frac{3}{7}$ and $\frac{2}{15}$, the numerator of $\frac{3}{7}$ and denominator of $\frac{2}{15}$ has common factor so, $\frac{3}{7} \times \frac{2}{15} = \frac{1}{7} \times \frac{2}{5} = \frac{2}{35}$

A Glimpse into Ancient Indian Mathematics

In ancient India, the process of simplifying fractions to its lowest form was called "Apavartana." This word means "simplifying by removing common factors."

The idea of Apavartana was so familiar in ancient times that it was mentioned even outside mathematics. Around 150 CE, a Jaina scholar Umasvati used the concept of Apavartana as an example in his philosophical texts. This shows that simplifying or "reducing" something to its purest form was such a well-known idea that it became a symbol of clarity and understanding, both in mathematics and philosophy.

ILLUSTRATIONS

4 A milkman has 12 litres of milk. He sells $\frac{4}{6}$ of it in the morning and $\frac{1}{4}$ of the total milk in the evening. The rest he keeps for his family. Find:

- (i) How much milk did he sell in total?
- (ii) How much milk is left for his family?

Soln.: Total quantity of milk = 12 litres

Milk sold in the morning = $\frac{4}{6}$ of 12 litres.

Firstly, let us simplify the fraction $\frac{4}{6}$ as its numerator and denominator has a common factor 2 so, $\frac{4}{6} = \frac{2}{3}$

Now, milk sold in the morning

$$= \frac{2}{3} \text{ of 12 litres}$$

$$= \frac{2}{3} \times 12 = 8 \text{ litres}$$

So, milk sold in the morning = 8 litres

Also, fraction of milk sold in the evening
 $= \frac{1}{4}$ of total milk.

$$\begin{aligned} \text{Milk sold in the evening} &= \frac{1}{4} \text{ of 12 litres} \\ &= \frac{1}{4} \times 12 = 3 \text{ litres} \end{aligned}$$

Milk sold in the evening = 3 litres

$$\begin{aligned} \text{(i) Total quantity of milk sold} &= 8 + 3 \\ &= 11 \text{ litres} \end{aligned}$$

Milk left = Total milk - Milk sold

$$= (12 - 11) \text{ litres} = 1 \text{ litre}$$

$$\text{(ii) Milk left for the family} = 1 \text{ litre}$$

5 What is $2\frac{1}{4}$ of $\frac{1}{18}$?

Soln.: The fraction $2\frac{1}{4}$ can be written as $\frac{9}{4}$

$$\text{Now, } \frac{9}{4} \text{ of } \frac{1}{18} = \frac{9}{4} \times \frac{1}{18} = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$$

Comparing Products in Fraction Multiplication

When we multiply two numbers, the product depends on whether the numbers are greater than 1 or between 0 and 1. So, there are three cases based on the number and their product. Let us understand them with the help of examples.

Case I: When both numbers are greater than 1 :

Example: The fraction $\frac{3}{2}$ and $\frac{5}{3}$ are greater than 1. Now, their product $= \frac{3}{2} \times \frac{5}{3} = \frac{5}{2}$, which is greater than both the numbers.

Case II: When both numbers are between 0 and 1 :

Example: The fractions $\frac{1}{2}$ and $\frac{1}{3}$ are lying between 0 and 1 and their product $= \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

The product is less than both the numbers.

Case III: When one number is between 0 and 1, the other is greater than 1:

Example: The fraction $\frac{2}{5}$ lies between 0 and 1 and number 5 is greater than 1. Now, their product $= \frac{2}{5} \times 5 = 2$

So, the product lies between the given two numbers.

So, here is the complete table based on three cases and the relationship between product and numbers.

Cases	Relationship between the product and the numbers
Both numbers greater than 1	Product is greater than the both numbers.
Both numbers between 0 and 1	Product is less than the both numbers.
One number between 0 and 1 and the other greater than 1	Product lies between the given two numbers

Note: (i) When we multiply a number by a fraction less than 1, the result becomes smaller. (It's like taking only a part of the number.)

(ii) When we multiply a number by a number greater than 1, the result becomes bigger. (It's like adding more to the number.)

Order of Multiplication

Suppose we need to find $\frac{1}{2}$ of $\frac{2}{5}$, then we find the product of $\frac{1}{2}$ and $\frac{2}{5}$ i.e., $\frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$.

Now, if we need to find $\frac{2}{5}$ of $\frac{1}{2}$, then also we need to find the product of $\frac{2}{5}$ and $\frac{1}{2}$ i.e., $\frac{2}{5} \times \frac{1}{2} = \frac{1}{5}$.

This implies $\frac{1}{2}$ of $\frac{2}{5}$ is same as $\frac{2}{5}$ of $\frac{1}{2}$.

It means that the order in which we multiply fractions does not matter. No matter which fraction comes first, the product remains the same.

Thus, for any two fractions $\frac{a}{b}$ and $\frac{x}{y}$, we have $\frac{a}{b} \times \frac{x}{y} = \frac{x}{y} \times \frac{a}{b}$

► For any two numbers a and b , $a \times b = b \times a$,

COMPETITION WINDOW



This is called the commutative property of multiplication.

ILLUSTRATIONS

6 Riya says, "When I multiply two fractions, I always get a smaller number." Is she correct? Explain your answer with examples for each case.

Soln.: Riya is not always correct. It depends on the types of fractions being multiplied. Let's look at all the possible cases to understand.

Case 1: Both numbers are between 0 and 1

For example; $\frac{1}{4} \times \frac{2}{5} = \frac{1}{10}$

Riya is correct in this case.

Case 2: Both numbers are greater than 1

For example; $\frac{5}{4} \times \frac{3}{2} = \frac{15}{8}$

$\frac{15}{8}$ is greater than both $\frac{5}{4}$ and $\frac{3}{2}$ so, Riya is wrong in this case, because the product is bigger.

Case 3: One number is between 0 and 1 and the other is greater than 1

For example; $\frac{1}{4} \times 4 = 1$

Here, 1 is greater than $\frac{1}{4}$ but smaller than 4

So, the product lies between the two numbers.

Riya is partly correct here as the product is smaller than one number but not both.

Reciprocal of a Fraction

Two non-zero fractions whose product with each other gives 1, are called reciprocals of each other.

For example; $\frac{3}{2} \times \frac{2}{3} = 1$

So, $\frac{2}{3}$ is reciprocal of $\frac{3}{2}$ and vice-versa.

Reciprocal can be obtained by interchanging numerator to denominator and denominator to numerator.

For example; For fraction $\frac{10}{3}$, Reciprocal = $\frac{3}{10}$

Note : Reciprocal of proper fraction is improper and vice-versa.

Division of Fractions

Division of a Whole Number by a Fraction

To divide a whole number by a fraction, multiply the whole number by reciprocal of the fraction.

For example; (i) $3 \div \frac{1}{4} = 3 \times \frac{4}{1} = 12$ (ii) $3 \div \frac{5}{2} = 3 \times \frac{2}{5} = \frac{6}{5}$

To divide a whole number by a mixed fraction, first convert the mixed fraction into improper fraction and then divide them.

Division of a Fraction by a Whole Number

To divide a fraction by a whole number, multiply the fraction by the reciprocal of the whole number.

Division of a Fraction by Another Fraction

To divide two fractions, we change the division sign into multiplication and write reciprocal of the divisor i.e., multiply fraction with the reciprocal of the divisor.

In general, if $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$

Note: Brahmagupta explained that “The division of fractions is performed by interchanging the numerator and denominator of the divisor; the numerator of the dividend is then multiplied by the (new) numerator, and the denominator by the (new) denominator.”

Bhāskara II (*Līlāvātī*, 1150 CE) described this as multiplying the first fraction by the reciprocal of the second.

ILLUSTRATIONS

7 Solve : $15 \div 3\frac{1}{2}$

Soln.: $15 \div 3\frac{1}{2} = 15 \div \frac{7}{2} = 15 \times \frac{2}{7} = \frac{30}{7} = 4\frac{2}{7}$

8 Simplify : $5\frac{1}{6} \div \frac{9}{2}$

Soln.: $5\frac{1}{6} \div \frac{9}{2} = \frac{31}{6} \div \frac{9}{2} = \frac{31}{6} \times \frac{2}{9} = \frac{31}{27} = 1\frac{4}{27}$

9 25 shirts of equal size were prepared from $30\frac{5}{9}$ m of cloth. Find the length of cloth used for making one shirt.

Soln.: Total length of cloth = $30\frac{5}{9}$ m = $\frac{275}{9}$ m

Number of shirts = 25

So, cloth required for making 1 shirt

$$= \left(\frac{275}{9} \div 25 \right) \text{ m}$$

$$= \left(\frac{275}{9} \times \frac{1}{25} \right) \text{ m} = \frac{11}{9} \text{ m} = 1\frac{2}{9} \text{ m}$$

Hence, $1\frac{2}{9}$ m cloth is used for making one shirt.

10 Jay cuts 20 m long rope into pieces of length $3\frac{1}{3}$ m each. How many pieces of rope did he get?

Soln.: Total length of the rope = 20 m

Length of 1 piece = $3\frac{1}{3}$ m = $\frac{10}{3}$ m

Number of pieces = $20 \div \frac{10}{3} = 20 \times \frac{3}{10} = 6$

Hence, Jay got 6 pieces of rope.

11 Product of two fractions is $\frac{3}{4}$. If one of the fraction is $7\frac{1}{2}$, then find the other.

Soln.: Product of two fractions = $\frac{3}{4}$

One of the fraction = $7\frac{1}{2} = \frac{15}{2}$

Other fraction = $\frac{3}{4} \div \frac{15}{2} = \frac{3}{4} \times \frac{2}{15} = \frac{1}{10}$

Hence, other fraction is $\frac{1}{10}$.

Comparing Quotient in Fraction Division

When we divide a number, the size of the quotient in comparison with dividend depends on whether the divisor is greater than 1 or less than 1.

Case I: When the divisor is greater than 1 :

(i) $\frac{2}{5} \div 2 = \frac{2}{5} \times \frac{1}{2} = \frac{1}{5} < \frac{2}{5}$

(ii) $2 \div \frac{5}{2} = 2 \times \frac{2}{5} = \frac{4}{5} < 2$

If the divisor is greater than 1, no matter the dividend is less than 1 or greater than 1, the quotient will always be smaller than the dividend.

Case II: When the divisor is less than 1:

(i) $\frac{2}{3} \div \frac{1}{2} = \frac{2}{3} \times 2 = \frac{4}{3} > \frac{2}{3}$

(ii) $2 \div \frac{2}{3} = 2 \times \frac{3}{2} = 3 > 2$

Whether the dividend is smaller or larger than 1, if the divisor is less than 1, the quotient will always be greater than the dividend.

Case III: When the divisor is equal to 1, then the dividend is same as quotient.

Fractional Relations — Finding a Part of a Part

Sometimes, we need to find what part of a figure or quantity is shaded, shared, or used. This part may itself be a fraction of another fraction or we can say a “part of a part.” To find such relations, we use fractional relations. To find how much of the whole it represents.

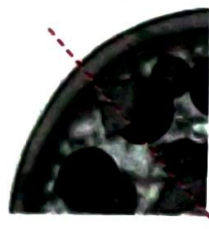
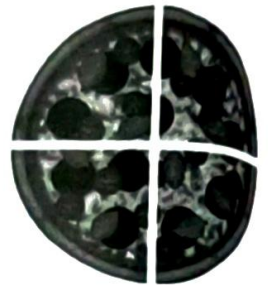
Let us understand this with the help of an example.

Suppose a pizza (a circle) is divided into 4 equal slices and you take $\frac{1}{4}$ of the whole pizza.

Now, your friend says she’ll eat $\frac{1}{2}$ of your slice.

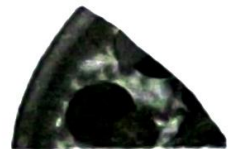
Now, we need to find how much of the whole pizza did your friend actually eat.

You had $\frac{1}{4}$ of the pizza and she ate $\frac{1}{2}$ of your $\frac{1}{4}$.



Amount of whole pizza ate by your friend = $\frac{1}{2}$ of $\frac{1}{4} = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

That means your friend ate $\frac{1}{8}$ of the whole pizza.



In general, if we take $\frac{a}{b}$ of the whole and then $\frac{c}{d}$ of that part will be $\frac{c}{d}$ of $\frac{a}{b} = \frac{c}{d} \times \frac{a}{b} = \frac{c \times a}{d \times b}$ of whole part.

ILLUSTRATIONS

12 Four painters are painting a wall. The first painter can paint the whole wall in $\frac{1}{6}$ of a day, the second can paint the wall in $\frac{1}{4}$ of a day, the third in $\frac{1}{2}$ of a day, and the fourth paint the wall in 1 day. If they all work together, how long will it take to paint a wall completely?

Soln.: The number of walls the first painter will paint in a day = $1 \div \frac{1}{6} = 6$

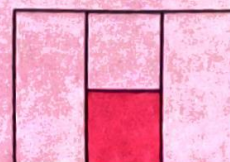
The second painter will paint
 $= 1 \div \frac{1}{4} = 4$ walls

The third painter will paint = $1 \div \frac{1}{2} = 2$ walls

The fourth painter will paint
 $= 1 \div 1 = 1$ wall

So, in one day, all four painters together can paint $6 + 4 + 2 + 1 = 13$ such walls. That means, in one day, they can complete 13 walls. So to paint 1 wall, they will need $1 \div 13 = \frac{1}{13}$ of a day.

13 What fraction of the whole rectangle is shaded?



Soln.: The rectangle is split into 3 equal vertical strips, so each strip is $\frac{1}{3}$ of the whole rectangle.

So, the middle strip = $\frac{1}{3}$

Now, the middle strip is divided into 2 equal

parts (top and bottom). One of these parts is shaded, so the shaded part = $\frac{1}{2}$ of the middle strip = $\frac{1}{2}$ of $\frac{1}{3}$ = $\frac{1}{2} \times \frac{1}{3}$ = $\frac{1}{6}$ of whole part.

So, the shaded region is $\frac{1}{6}$ of the whole rectangle.

Fractions in Ancient India

Bhāskarāchārya's in his famous book *Līlāvati* (1150 CE), presented a humorous problem about a miser gave a beggar a very tiny fraction $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{1}{5} \times \frac{1}{16} \times \frac{1}{4}$ of a dramma (a silver coin). The fraction, when evaluated, equals $\frac{1}{1280}$, which corresponds to one cowrie shell as

1 Dramma = 1280 cowrie shells. This humorous problem illustrates how the miser, despite giving what seems like a large amount, only offers the beggar the smallest unit of money. The text also provides insight into the coinage system of the time, where gold dinars, silver drammās, copper panas, and cowrie shells were used in transactions, with cowrie shells being the lowest denomination.

Thus, fractions played a key role in Ancient Indian Mathematics, with early used in texts like the *Śulbasūtras* (800 BCE) and Jain works (150 BCE). Brahmagupta's *Brāhmasphuṭasiddhānta* (628 CE) formalized rules for fraction operations, while Bhāskara II refined these in his *Līlāvati* (1150 CE), introducing the concept of reciprocals for division. Indian mathematicians such as Śhrīdharāchārya and Bhāskara II contributed significantly to the theory of fractions, which was later transmitted to the Arab world and Europe, shaping modern arithmetic.

ILLUSTRATIONS

14 If 1 copper pana = 30 cowrie shells, then what part of a pana is 6 cowrie shells?

Soln.: 1 copper pana = 30 cowrie shells

So, 1 cowrie shell = $\frac{1}{30}$ of copper pana

Hence, 6 cowrie shells = $\frac{6}{30}$ of copper pana
= $\frac{1}{5}$ of copper pana.

CONCEPT MAP

WORKING WITH FRACTIONS

MULTIPLICATION OF FRACTION

DIVISION OF FRACTION

MULTIPLICATION OF A FRACTION BY A WHOLE NUMBER

To multiply a fraction by a whole number, multiply the numerator of the fraction with that whole number keeping the denominator same.

MULTIPLICATION OF A FRACTION BY A FRACTION

To multiply a fraction by a fraction, multiply their numerators and denominators.

$$\text{Product of two fractions} = \frac{\text{Product of numerators}}{\text{Product of denominators}}$$

$$\text{If } \frac{a}{b} \text{ and } \frac{c}{d} \text{ are two fractions, then } \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

MULTIPLICATION OF FRACTIONS USING AREA MODEL

The area model visually demonstrates multiplying fractions by dividing a rectangle into equal parts, showing how one fraction of another result in smaller part of the whole. For example, $1/2 \times 1/3$ equals $1/6$.

MULTIPLICATION OF FRACTIONS- SIMPLIFYING TO LOWEST FORM

When we multiply fractions, we can make multiplication easier by simplifying the fraction that means dividing the numerator and denominator by their common factors to reduce the fraction to its simplest form.

A GLIMPSE INTO ANCIENT INDIAN MATHEMATICS

In ancient India, the process of reducing a fraction to its lowest terms was called Apavartana. This word means "simplifying by removing common factors."

RECIPROCAL OF A FRACTION

Two non-zero fractions whose product with each other gives 1, are called reciprocals of each other. Reciprocal can be obtained by interchanging numerator to denominator and denominator to numerator. Reciprocal of proper fraction is improper and vice-versa.

Division of a Whole Number by a Fraction

To divide a whole number by a fraction, multiply the whole number by reciprocal of the fraction.

Division of a Fraction by a Whole Number

To divide a fraction by a whole number, multiply the fraction by the reciprocal of the whole number.

Division of a Fraction by Another Fraction

To divide two fractions, multiply fraction with the reciprocal of the divisor.

In general, if $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

COMPARING QUOTIENT IN FRACTION DIVISION

Case I: If the divisor is greater than 1, no matter the dividend is less than 1 or greater than 1, the quotient will always be smaller than the dividend.

Case II: Whether the dividend is smaller or larger than 1, if the divisor is less than 1, the quotient will always be greater than the dividend.

Case III: When the divisor is equal to 1, then the dividend is same as quotient.

COMPARING PRODUCTS IN FRACTION MULTIPLICATION

Cases	Relationship between the product and the numbers
Both numbers greater than 1	Product is greater than the both numbers.
Both numbers between 0 and 1	Product is less than the both numbers.
One number between 0 and 1 and the other greater than 1	Product lies between the given two numbers

Solved Examples

1. Find :

- (i) $\frac{1}{4}$ of a rupee (ii) $\frac{3}{4}$ of a day
(iii) $\frac{7}{25}$ of a kg (iv) $\frac{2}{3}$ of an hour.

Soln.: (i) $\frac{1}{4}$ of a rupee = $\frac{1}{4}$ of 100 paise
 $= \frac{1}{4} \times 100$ paise = 25 paise

(ii) $\frac{3}{4}$ of a day = $\frac{3}{4}$ of 24 hours
 $= \frac{3}{4} \times 24 = 18$ hours

(iii) $\frac{7}{25}$ of a kg = $\frac{7}{25}$ of 1000 gm
 $= \frac{7}{25} \times 1000$ gm = 280 gm

(iv) $\frac{2}{3}$ of an hour = $\frac{2}{3}$ of 60 minutes
 $= \frac{2}{3} \times 60$ minutes = 40 minutes

2. Which is greater : $\frac{3}{4}$ of 36 or $\frac{4}{5}$ of 35?

Soln.: $\frac{3}{4}$ of 36 = $\frac{3}{4} \times 36 = 27$ and

$\frac{4}{5}$ of 35 = $\frac{4}{5} \times 35 = 28$

Since, $28 > 27$ $\therefore \frac{4}{5}$ of 35 > $\frac{3}{4}$ of 36

3. Shikha's age is $\frac{2}{5}$ of the age of her eldest brother. If her brother's age is 35 years, then how old is Shikha?

Soln.: Shikha's age = $\frac{2}{5}$ of the age of her brother
 $= \frac{2}{5}$ of 35 years = $\frac{2}{5} \times 35$ years = 14 years

So, Shikha is 14 years old.

4. Vinay covers $7\frac{1}{2}$ km in one hour. How much distance will he cover in $2\frac{4}{5}$ hours?

Soln.: $7\frac{1}{2}$ km = $\frac{15}{2}$ km and $2\frac{4}{5}$ hours = $\frac{14}{5}$ hours

Distance covered in one hour = $\frac{15}{2}$ km

So, distance covered in $\frac{14}{5}$ hours = $\frac{15}{2} \times \frac{14}{5}$ km
 $= 21$ km

Thus, Vinay will cover 21 km in $2\frac{4}{5}$ hours.

5. If the cost of 17 m of cloth is ₹ $77\frac{5}{7}$. Find its cost per metre.

Soln.: Cost of 17 m of cloth = ₹ $77\frac{5}{7}$ = ₹ $\frac{544}{7}$

Cost of 1 m of cloth = $\frac{544}{7} \div 17 = \frac{544}{7} \times \frac{1}{17}$
 $= \frac{32}{7} = 4\frac{4}{7}$

Hence, the cost of cloth is ₹ $4\frac{4}{7}$ per metre.

6. Multiply $\frac{25}{4}$ and $\frac{2}{7}$. Compare the product with both fractions. Is the product smaller than both fractions, greater than both, or lies in between? Explain why.

Soln.: $\frac{25}{4} \times \frac{2}{7} = \frac{25 \times 2}{4 \times 7} = \frac{25}{14} > 1$

Now, $\frac{25}{4} > \frac{25}{14}$ and $\frac{2}{7} = \frac{4}{14} < \frac{25}{14}$

So, the product lies between the given fractions because one fraction is greater than 1 and the other is between 0 and 1.

7. Find the product of the following expressions and observe what happens each time:

(i) $\left(1 + \frac{1}{2}\right)$

(ii) $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)$

(iii) $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)$

$$(iv) \left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right)$$

Write what you notice about the numerators and denominators of the products.

Soln: (i) $\left(1 + \frac{1}{2}\right) = \frac{3}{2}$

(ii) $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right) = \left(\frac{3}{2}\right) \times \left(\frac{4}{3}\right) = \frac{12}{6} = 2 = \frac{4}{2}$

(iii) $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right) = \left(\frac{3}{2}\right) \times \left(\frac{4}{3}\right) \times \left(\frac{5}{4}\right) = \frac{5}{2}$

(iv) $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right)$
 $= \left(\frac{3}{2}\right) \times \left(\frac{4}{3}\right) \times \left(\frac{5}{4}\right) \times \left(\frac{6}{5}\right) = \frac{6}{2}$

So, each time we include a new term, the numerator increases by 1, while the denominator remains the same i.e., 2.

8. A goldsmith decided to donate a part of his gold coin collection to a temple. He gave $\frac{1}{2}$ of $\frac{1}{3}$ of $\frac{1}{4}$ of $\frac{1}{5}$ of his total coins. If he originally had 600 coins, how many coins did he donate?

Soln: The goldsmith donates $\frac{1}{2}$ of $\frac{1}{3}$ of $\frac{1}{4}$ of $\frac{1}{5}$
 i.e., $\frac{1}{2} \times \frac{1}{3} \times \frac{1}{4} \times \frac{1}{5} = \frac{1}{120}$ of his total coins.

Now, the number of coins donated = $\frac{1}{120}$ of
 $600 = \frac{1}{120} \times 600 = 5$

So, the goldsmith donated 5 gold coins to the temple.

9. A milkman has $\frac{3}{4}$ litres of milk. He wants to fill bottles that each hold $\frac{1}{8}$ litre of milk. How many bottles can he fill?

Soln: Quantity of milk available = $\frac{3}{4}$ litres

Bottle capacity = $\frac{1}{8}$ litre

Number of bottles can be filled

$$= \frac{\text{Quantity of milk available}}{\text{Capacity of Bottle}}$$

$$= \frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times 8 = 6$$

Thus, he can fill 6 bottles.

10. A baker has $2\frac{1}{2}$ kg of cake. He wants to cut it into pieces, each weighing $\frac{1}{4}$ kg. How many pieces of cake will he get?

Soln: Quantity of cake = $2\frac{1}{2}$ kg = $\frac{5}{2}$ kg

Weight of one piece of cake = $\frac{1}{4}$ kg

So, number of pieces = $\frac{\text{Quantity of cake}}{\text{Weight of one piece}}$

$$= \frac{5}{2} \div \frac{1}{4} = \frac{5}{2} \times 4 = \frac{20}{2} = 10$$

So, the baker will get 10 pieces of cake.

11. A class of 30 students is asked to visit three sections of a library. They first split equally into two groups. One group goes to the Science section. The other group splits again, with $\frac{1}{3}$ going to Arts section and $\frac{2}{3}$ going to History section. How many students visit each section?

Soln: Total students in a class = 30

The class split equally into two groups:

So, number of students in each group = $\frac{1}{2}$ of 30
 $= \frac{1}{2} \times 30 = 15$

So, 15 students go to the Science section and other group of 15 students splits as $\frac{1}{3}$ going to Arts section and $\frac{2}{3}$ going to History section.

Number of students go to Art section = $\frac{1}{3}$ of 15
 $= \frac{1}{3} \times 15 = 5$ students.

Number of students go to History section
 $= \frac{2}{3}$ of 15 = $\frac{2}{3} \times 15 = 10$ students.

12. Avinash had ₹ 55,000 with him. He kept $\frac{1}{11}$ of it for himself and gave half of remaining money to his wife and divide the rest money equally among his four children. How much money did each child get?

Soln: Total amount with Avinash = ₹ 55,000

Amount that Avinash kept for himself

$$= ₹ \left(\frac{1}{11} \text{ of } 55,000 \right) = ₹ 5000$$

Remaining amount = $55,000 - 5,000 = ₹ 50,000$

Amount received by Avinash's wife

$$= ₹ \left(\frac{1}{2} \times 50,000 \right) = ₹ 25,000$$

Amount left for his children

$$= ₹ (50,000 - 25,000)$$

$$= ₹ 25,000$$

Amount received by each child = ₹ $(25,000 \div 4)$

$$= ₹ \frac{25000}{4} = ₹ 6250$$

13. Out of 140 mangoes, a man sells half of the number of existing mangoes plus 1 to the first customer, then sells $\frac{1}{3}$ of the remaining number of mangoes plus 1 to the second customer, then $\frac{1}{5}$ of the remaining number of mangoes plus 1 to the third customer and $\frac{1}{7}$ of the remaining number of mangoes plus 1 to the fourth customer. How many mangoes are left unsold?

Soln: Total number of mangoes = 140

Mangoes sold to first customer

$$= \frac{1}{2} \times 140 + 1 = 70 + 1 = 71$$

Remaining mangoes = $140 - 71 = 69$

Mangoes sold to second customer

$$= \frac{1}{3} \times 69 + 1 = 23 + 1 = 24$$

Now, remaining mangoes = $69 - 24 = 45$

$$\begin{aligned} \text{Mangoes sold to third customer} &= \frac{1}{5} \times 45 + 1 \\ &= 9 + 1 = 10 \end{aligned}$$

Again, remaining mangoes = $45 - 10 = 35$

$$\begin{aligned} \text{Mangoes sold to fourth customer} &= \frac{1}{7} \times 35 + 1 \\ &= 5 + 1 = 6 \end{aligned}$$

Number of mangoes left unsold = $35 - 6 = 29$

14. In a class of 63 students $\frac{1}{7}$ of the total students like to play football, $\frac{2}{7}$ of the total like to play Hockey and the remaining students like to play Kho Kho.

(i) How many students like to play football?

(ii) How many students like to play Hockey?

(iii) What fraction of the total number of students like to play Kho Kho?

Soln: Total number of students in the class = 63

(i) Number of students who like to play

$$\text{Football} = \frac{1}{7} \text{ of } 63 = \frac{1}{7} \times 63 = 9$$

(ii) Number of students who like to play

$$\text{Hockey} = \frac{2}{7} \text{ of } 63 = \frac{2}{7} \times 63 = 18$$

(iii) Number of students who like to play

$$\text{Kho Kho} = 63 - (9 + 18) = 36$$

∴ Fraction of students who like to play Kho Kho

$$= \frac{36}{63} = \frac{4}{7}$$

15. Kavita has bought a carpet of size

$$3 \text{ m} \times 5\frac{5}{3} \text{ m. But her room size is } 3\frac{3}{2} \text{ m} \times 2\frac{4}{5} \text{ m.}$$

Find the area to be cut off to fill wall to wall carpet into the room?

$$\text{Soln: Area of carpet} = \left(3 \times 5\frac{5}{3} \right) \text{ m}^2$$

$$= \frac{3 \times 20}{3} = 20 \text{ m}^2$$

$$\text{Area of room} = \left(3\frac{3}{2} \times 2\frac{4}{5} \right) \text{ m}^2 = \frac{126}{10} = 12.6 \text{ m}^2$$

∴ Area to be cut off

$$= (20 - 12.6) \text{ m}^2 = 7.4 \text{ m}^2 = \left(\frac{74}{10} \right) \text{ m}^2$$

NCERT Section

Figure it Out

(Page 176 & 177)

1. Tenzin drinks $\frac{1}{2}$ glass of milk every day. How many glasses of milk does he drink in a week? How many glasses of milk did he drink in the month of January?

2. A team of workers can make 1 km of a water canal in 8 days. So, in one day, the team can make _____ km of the water canal. If they work 5 days a week, they can make _____ km of the water canal in a week.

3. Manju and two of her neighbours buy 5 litres of oil every week and share it equally among the 3 families. How much oil does each family get in a week? How much oil will one family get in 4 weeks?

4. Safia saw the Moon setting on Monday at 10 pm. Her mother, who is a scientist, told her that every day the Moon sets $\frac{5}{6}$ hour later than the previous day. How many hours after 10 pm will the moon set on Thursday?

5. Multiply and then convert it into a mixed fraction:

(a) $7 \times \frac{3}{5}$

(b) $4 \times \frac{1}{3}$

(c) $\frac{9}{7} \times 6$

(d) $\frac{13}{11} \times 6$

Figure it Out

(Page 180 & 181)

1. Find the following products. Use a unit square as a whole for representing the fractions:

(a) $\frac{1}{3} \times \frac{1}{5}$

(b) $\frac{1}{4} \times \frac{1}{3}$

(c) $\frac{1}{5} \times \frac{1}{2}$

(d) $\frac{1}{6} \times \frac{1}{5}$

2. Find the following products. Use a unit square as a whole for representing the fractions and carrying out the operations.

(a) $\frac{2}{3} \times \frac{4}{5}$

(b) $\frac{1}{4} \times \frac{2}{3}$

(c) $\frac{3}{5} \times \frac{1}{2}$

(d) $\frac{4}{6} \times \frac{3}{5}$

Figure it Out

(Page 183 & 184)

1. A water tank is filled from a tap. If the tap is open for 1 hour, $\frac{7}{10}$ of the tank gets filled. How much of the tank is filled if the tap is open for

(a) $\frac{1}{3}$ hour _____

(c) $\frac{2}{3}$ hour _____

(b) $\frac{3}{4}$ hour _____

(d) $\frac{7}{10}$ hour _____

(e) For the tank to be full, how long should the tap be running?

2. The government has taken $\frac{1}{6}$ of Somu's land to build a road. What part of the land remains with Somu now? She gives half of the remaining part of the land to her daughter, Krishna, and $\frac{1}{3}$ it to her son Bora. After giving them their shares, she kept the remaining land for herself.

(a) What part of the original land did Krishna get?

(b) What part of the original land did Bora get?

(c) What part of the original land did Somu keep for herself?

3. Find the area of a rectangle of sides $3\frac{3}{4}$ ft and $9\frac{3}{5}$ ft.

4. Tsewang plants four saplings in a row in his garden. The distance between two saplings is $\frac{3}{4}$ m. Find the distance between the first and last sapling.

[Hint: Draw a rough diagram with four saplings with distance between two saplings as $\frac{3}{4}$ m]

5. Which is heavier: $\frac{12}{15}$ of 500 grams or $\frac{3}{20}$ of 4 kg?

Figure it Out

(Page 196)

1. Evaluate the following :

$3\frac{7}{9}$	$\frac{14}{4} \div 2$	$\frac{2}{3} \div \frac{2}{3}$	$\frac{14}{6} \div \frac{7}{3}$
$\frac{4}{3} \div \frac{3}{4}$	$\frac{7}{4} \div \frac{1}{7}$	$\frac{8}{2} \div \frac{4}{15}$	
$\frac{1}{5} \div \frac{1}{9}$	$\frac{1}{6} \div \frac{11}{12}$	$3\frac{2}{3} \div 1\frac{3}{8}$	

2. For each of the questions below, choose the expression that describes the solution. Then simplify it.

- (a) Maria bought 8 m of lace to decorate the bags she made for school. She used $\frac{1}{4}$ m for each bag and finished the lace. How many bags did she decorate?

(i) $8 \times \frac{1}{4}$ (ii) $\frac{1}{8} \times \frac{1}{4}$

(iii) $8 \div \left(\frac{1}{4}\right)$ (iv) $\frac{1}{4} \div 8$

- (b) $\frac{1}{2}$ meter of ribbon is used to make 8 badges. What is the length of the ribbon used for each badge?

(i) $8 \times \frac{1}{2}$ (ii) $\frac{1}{2} \div \left(\frac{1}{8}\right)$

(iii) $8 \div \left(\frac{1}{2}\right)$ (iv) $\frac{1}{2} \div 8$

- (c) A baker needs $\frac{1}{6}$ kg of flour to make one loaf of bread. He has 5 kg of flour. How many loaves of bread can he make?

(i) $5 \times \frac{1}{6}$ (ii) $\frac{1}{6} \div 5$

(iii) $5 \div \left(\frac{1}{6}\right)$ (iv) 5×6

3. If $\frac{1}{4}$ kg of flour is used to make 12 rotis, how much flour is used to make 6 rotis?

4. *Pāṭiganita*, a book written by Sridharacharya in the 9th century CE, mentions this problem: "Friend, after thinking, what sum will be obtained by adding together $1 \div \frac{1}{6}, 1 \div \frac{1}{10}, 1 \div \frac{1}{13}, 1 \div \frac{1}{9}, 1 \div \frac{1}{2}$." What should the friend say?

5. Mira is reading a novel that has 400 pages. She read $\frac{1}{5}$ of the pages yesterday and $\frac{3}{10}$ of the pages today. How many more pages does she need to read to finish the novel?

6. A car runs 16 km using 1 litre of petrol. How far will it go using $2\frac{3}{4}$ litres of petrol?

7. Amritpal decides on a destination for his vacation. If he takes a train, it will take him $5\frac{1}{6}$ hours to get there. If he takes a plane, it will take him $\frac{1}{2}$ hour. How many hours does the plane save?

8. Mariam's grandmother baked a cake. Mariam and her cousins finished $\frac{4}{5}$ of the cake. The remaining cake was shared equally by Mariam's three friends. How much of the cake did each friend get?

9. Choose the option(s) describing the product of $\left(\frac{565}{465} \times \frac{707}{676}\right)$:

- (a) $> \frac{565}{465}$ (b) $< \frac{565}{465}$
 (c) $> \frac{707}{676}$ (d) $< \frac{707}{676}$
 (e) > 1 (f) < 1

10. What fraction of the whole square is shaded?



11. A colony of ants set out in search of food. As they search, they keep splitting equally at each point (as shown in the Fig. 8.7) and reach two food sources, one near a mango tree and another near a sugarcane field. What fraction of the original group reached each food source?



12. What is $1 - \frac{1}{2}$?

$$\left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right)?$$

$$\left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{4}\right) \times \left(1 - \frac{1}{5}\right)?$$

$$\left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{4}\right) \times \left(1 - \frac{1}{5}\right) \times \left(1 - \frac{1}{6}\right) \times \left(1 - \frac{1}{7}\right) \times \left(1 - \frac{1}{8}\right) \times \left(1 - \frac{1}{9}\right) \times \left(1 - \frac{1}{10}\right)?$$

Make a general statement and explain.

Exercise

Multiple Choice Questions

LEVEL - 1

Multiplication of Fractions

1. A farmer has 192 animals. $\frac{7}{16}$ of animals are cattle. $\frac{2}{3}$ of cattle are dairy cows. How many dairy cows he has?
 (a) 128 (b) 84
 (c) 56 (d) 112

(NCERT Page 173) **An**

2. A jar has 40 marbles, $\frac{2}{5}$ of them are red and rest are blue, the number of blue marbles is

(a) 16

(c) 24

(b) 32

(d) 12 (Page 173) **Ev**

3. What is $3\frac{1}{2}$ of 20?

(a) 70

(c) 50

(b) 60

(d) 40 (Page 173) **Ev**

4. Soni and Ravi went for a picnic. Their mother gave them a water bottle that contained 5 litres of water. Soni consumed $\frac{2}{5}$ of the water. Ravi consumed the remaining water. How much water did Ravi drink?

(a) 3 litres

(b) 2 litres

(c) 1 litre

(d) $\frac{1}{3}$ litre

(Page 173)

5. A can has $\frac{5}{6}$ litre of oil. If $\frac{2}{5}$ of it spills, how much oil is lost?

(a) $\frac{1}{2}$ litre (b) $\frac{1}{3}$ litre
(c) $\frac{1}{6}$ litre (d) $\frac{2}{3}$ litre

(Page 177) **Ap**

6. A farmer harvested $\frac{1}{2}$ part of grains from his field. He sold $\frac{1}{2}$ of what he harvested. What fraction of the whole grain in his field did he sell?

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{8}$ (d) $\frac{1}{16}$

Ap

7. In the area model for $\frac{1}{5} \times \frac{1}{6}$, how many small equal parts will the whole be divided into?

(a) 11 (b) 30 (c) 1 (d) 2

(Page 180) **U**

8. The length of rectangle is $3\frac{1}{2}$ m and its breadth is $1\frac{3}{4}$ m. Find its area.

(a) $6\frac{1}{8}$ sq. m (b) $5\frac{5}{8}$ sq. m
(c) $6\frac{1}{4}$ sq. m (d) $7\frac{1}{8}$ sq. m (Page 180)

9. The value of $\frac{12}{15} \times \frac{10}{18}$ is

(a) $\frac{2}{9}$ (b) $\frac{2}{3}$
(c) $\frac{4}{9}$ (d) $\frac{3}{4}$

(Page 182) **U**

10. What fraction of the whole rectangle is shaded?



(a) $\frac{2}{3}$
(c) $\frac{1}{6}$

(b) $\frac{1}{8}$
(d) $\frac{3}{8}$

(Page 182) **U**

11. If one fraction is less than 1 and other is greater than 1, then their product is
(a) greater than both the numbers.
(b) smaller than both the numbers.
(c) lies between both numbers.
(d) can't be determined. (Page 184) **R**

12. Choose the option describing the product of $\frac{176}{681} \times \frac{126}{421}$

(a) $> \frac{126}{421}$

(b) $> \frac{176}{681}$

(c) > 1

(d) $< \frac{176}{681}$

(Page 184) **U**

13. For any two fractions $\frac{a}{b}$ and $\frac{c}{d}$, choose the correct statement.

(a) $\frac{a}{b} \times \frac{c}{d} > \frac{c}{d} \times \frac{a}{b}$

(b) $\frac{a}{b} \times \frac{c}{d} < \frac{c}{d} \times \frac{a}{b}$

(c) $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$

(d) None of these

(Page 186) **R**

14. A pizza is cut into 4 equal slices. Ayaan eat $\frac{1}{2}$ of a slice. How much of the whole pizza did you eat?

(a) $\frac{1}{3}$

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) $\frac{1}{8}$

Ap

Division of Fractions

15. A truck carries $\frac{5}{6}$ tonne of sand per trip. How many trips are needed to carry 5 tonnes of sand?

(a) 3

(b) 4

(c) 5

(d) 6

(Page 186) **Ev**

16. The product of two numbers is $\frac{28}{27}$. If one of the number is $\frac{5}{9}$, find the other.

(a) $\frac{28}{15}$ (b) $\frac{29}{15}$
(c) $\frac{9}{5}$ (d) $\frac{28}{9}$ (Page 186) **An**

17. Reciprocal of $1\frac{3}{4}$ is

(a) $1\frac{4}{3}$ (b) $4\frac{1}{3}$
(c) $3\frac{1}{4}$ (d) None of these

(Page 188) **U**

18. Reciprocal of $\frac{5}{4} \times \frac{7}{3}$ is

(a) $\frac{5}{4} \times \frac{7}{3}$ (b) $\frac{4}{5} \times \frac{3}{7}$
(c) $\frac{4}{5} \times \frac{7}{3}$ (d) $\frac{5}{4} \times \frac{3}{7}$

(Page 188) **U**

19. When divisor is equal to 1, then dividend is

(a) greater than quotient.
(b) equal to quotient.
(c) smaller than quotient.
(d) can't be determined.

(Page 189) **R**

20. The perimeter of a square is $43\frac{1}{6}$ cm. Find the length of each side of the square.

(a) $172\frac{2}{3}$ cm (b) $10\frac{19}{24}$ cm
(c) $64\frac{1}{2}$ cm (d) $10\frac{9}{24}$ cm

(Page 190) **Ap**

21. By selling tickets for ₹ $52\frac{1}{2}$ each, the money collected was ₹ $5512\frac{1}{2}$. Find the number of tickets sold.

(a) 105 (b) 289
(c) 106 (d) 546 (Page 190) **Ev**

22. Meena has $\frac{3}{5}$ litre of paint. Each wall needs $\frac{1}{10}$ litre. How many walls can she paint?

(a) 6
(c) 4

(b) 5
(d) 3 (Page 190) **Ev**

23. A cyclist covers $\frac{5}{6}$ km in $\frac{1}{9}$ hour. What distance will he cover in 3 hours?

(a) $\frac{45}{2}$ km (b) 10 km
(c) 15 km (d) $\frac{5}{18}$ km

(Page 190) **Ap**

24. The expression $\frac{1}{15} \div \left(\frac{4}{15} + \frac{1}{3} \right)$ is equivalent to

(a) $\frac{1}{9}$ (b) 9
(c) $\frac{1}{5}$ (d) 5

Ev

25. The sum obtained by adding the given numbers together is

$1\frac{1}{4}, 1\frac{1}{6}, 1\frac{1}{17}, 1\frac{1}{21}$

(a) $\frac{1}{24}$ (b) 24
(c) $\frac{1}{48}$ (d) 48

Ev

LEVEL - 2

26. From a wire of length $10\frac{1}{2}$ metres long

Samir cut off 5 pieces of length $1\frac{1}{4}$ metres each. How much length of the wire was left?

(a) $4\frac{1}{4}$ m (b) $21\frac{1}{4}$ m
(c) $6\frac{1}{4}$ m (d) $5\frac{1}{4}$ m

Ev

27. $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \dots \left(1 - \frac{1}{10}\right) =$

(a) $\frac{10}{11}$ (b) $\frac{1}{9}$
(c) $\frac{1}{10}$ (d) $\frac{1}{2}$

Ev

28. Find the reciprocal of $\frac{3}{4} \times \left(5\frac{1}{3} + 2\frac{1}{5}\right) + \frac{7}{9}$.

- (a) $1\frac{59}{140}$ (b) $4\frac{59}{140}$ (c) $3\frac{59}{140}$ (d) $7\frac{37}{140}$

29. The product of a fraction and the sum of $3\frac{1}{5}$ and $3\frac{1}{10}$ is $\frac{9}{10}$. Find the fraction.

- (a) $5\frac{7}{10}$ (b) $\frac{9}{70}$
(c) $\frac{1}{7}$ (d) 7

Ev

30. Find the reciprocal of the sum of $1\frac{7}{9}$ and $1\frac{2}{7}$.

- (a) $3\frac{4}{63}$ (b) $\frac{63}{256}$ (c) $4\frac{4}{63}$ (d) $\frac{63}{193}$

31. Shreya plants 5 saplings, in a row, in her garden. The distance between two adjacent saplings is $\frac{5}{4}$ m. Find the distance between the first and the last sapling.

- (a) 4 m (b) $4\frac{1}{2}$ m
(c) 5 m (d) $5\frac{1}{4}$ m

Ev

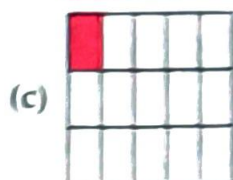
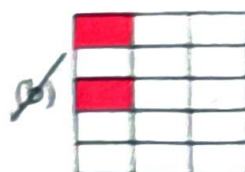
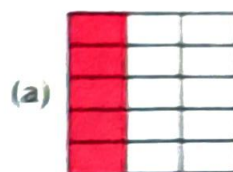
32. A car runs 20 km using 1 litre of petrol. How much distance will it cover using $2\frac{3}{4}$ litres of petrol?

- (a) 44 km (b) 34 km
(c) 48 km (d) 55 km

33. Lipika reads a book for $1\frac{3}{4}$ hours everyday. She reads the entire book in 6 days. How many hours in all were required by her to read the book?

- (a) $5\frac{1}{2}$ (b) $7\frac{1}{2}$
(c) $8\frac{1}{2}$ (d) $10\frac{1}{2}$

34. Which of the given figure represents a shaded part as $\frac{2}{5}$ of $\frac{1}{3}$?



(d) None of these

Cr

35. If 1 gold dinar = 12 silver drammass
1 silver dramma = 4 copper panas,
then what fraction of gold dinar is 1 copper pana?

- (a) $\frac{1}{12}$ (b) $\frac{1}{48}$
(c) $\frac{1}{18}$ (d) $\frac{1}{36}$

Ap

36. Find the simplified value of the given product of fractions for $k = 2$.

$$\frac{k}{k+1} \times \frac{k+1}{k+2} \times \frac{k+2}{k+3} \times \frac{k+3}{k+4}$$

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) 1

Ev

37. A river splits into two branches. One of those again splits into 4 smaller streams, what fraction of the river's water flows in each of the smaller streams?

- (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{1}{6}$ (d) $\frac{1}{7}$

38. A rope is $6\frac{3}{4}$ metres long. It is cut into pieces each $\frac{1}{4}$ metre long. How many pieces will be obtained?

- (a) 18 (b) 24 (c) 27 (d) 30

39. What is $\frac{1}{4}$ of $2\frac{1}{7}$ of $\frac{5}{2}$?

- (a) $1\frac{19}{56}$ (b) $1\frac{19}{28}$
(c) $\frac{15}{28}$ (d) None of these

Ev

40. A worker completes $\frac{2}{3}$ of a job in 4 hours. How long will it take him to complete the entire job at the same pace?

(a) 5 hours (b) $5\frac{1}{2}$ hours
(c) 6 hours (d) $6\frac{1}{2}$ hours

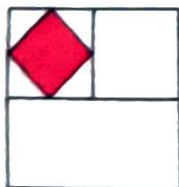
41. If $\frac{2}{3}$ of a number equals $\frac{4}{5}$, what is the number?

(a) $\frac{3}{5}$ (b) $\frac{6}{5}$
(c) $\frac{7}{5}$ (d) $\frac{9}{10}$

42. A pot contains 64 litres of milk. A cat drinks $\frac{1}{4}$ of $\frac{1}{2}$ of $\frac{3}{8}$ of it. How much milk did the cat drink?

(a) 3 litres (b) 4 litres
(c) 5 litres (d) 6 litres

43. In the given figure, find the fraction of the big square that the shaded region occupies.



$\frac{6}{16}$



(a) $\frac{1}{16}$ (b) $\frac{1}{12}$
(c) $\frac{1}{10}$ (d) $\frac{1}{8}$

44. A company gives $\frac{2}{5}$ of its profit to charity. The charity spends $\frac{3}{8}$ of that on education. What fraction of the company's total profit goes to education?

(a) $\frac{3}{8}$ (b) $\frac{6}{8}$
(c) $\frac{3}{20}$ (d) $\frac{7}{20}$

45. Three workers can do a job in 4 days, 6 days and 12 days. If they working together, then they will finish the job in

(a) 2 days (b) $2\frac{2}{3}$ days
(c) $1\frac{1}{2}$ days (d) 1 day

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

46. Select the correct match.

Figure

Fraction of unshaded part



$\frac{2}{9}$



$\frac{2}{3}$



$\frac{1}{6}$



$\frac{2}{3}$

47. Arrange the shaded fraction of following figures in ascending order.



P



Q



R



S

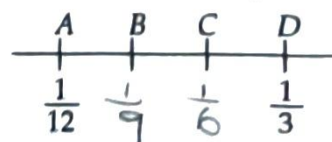
(a) P, Q, R, S
(c) Q, R, P, S

(b) S, Q, R, P
(d) R, S, Q, P

48. How many minutes are there in $1\frac{3}{4}$ hours?

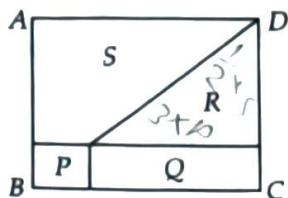
(a) 75 (b) 105
(c) 115 (d) 95

49. The number line below shows 4 fractions, A, B, C and D. If $AB = BC = CD$, then find the values of B and C respectively.



(a) $\frac{1}{6}, \frac{1}{4}$ (b) $\frac{1}{4}, \frac{1}{6}$ (c) $\frac{1}{5}, \frac{1}{6}$ (d) $\frac{1}{6}, \frac{1}{5}$

50. The figure below shows the rectangle ABCD (not drawn to scale). R is thrice the size of P and half the size of S. Q is twice the size of P. What fraction of the whole figure is the sum of Q and S?



- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{4}{5}$ (d) $\frac{6}{5}$
51. How many $\frac{1}{4}$ are there in $22\frac{1}{2}$?

- (a) 22 (b) 44
(c) 45 (d) 90

52. How many more unit squares in the figure must be shaded so that the fraction of unshaded squares becomes $\frac{2}{5}$?



- (a) 5 (b) 6
(c) 3 (d) 2

53. Mr. Sharma earns ₹ 64000 per month. He gave $\frac{2}{5}$ of his salary to his wife and $\frac{5}{8}$ of the remaining to his daughter and the rest to his son. Find the share of his son.

- (a) ₹ 14400 (b) ₹ 24000
(c) ₹ 25600 (d) ₹ 38400

54. Evaluate :
$$\frac{\left(1 + \frac{2}{9}\right) - \left[2\frac{1}{2} - \left(1\frac{1}{2} - \left(\frac{-1}{3}\right)\right)\right]}{\left(1 + 3\frac{1}{5}\right) + \left(1 + 2\frac{2}{3}\right)}$$

- (a) $5\frac{19}{33}$ (b) $5\frac{13}{19}$
(c) $3\frac{8}{33}$ (d) $13\frac{5}{6}$

55. In a hall, $\frac{1}{3}$ of the people were females and $\frac{1}{2}$ of those females were school girls. If the total number of people were 600, then how many school girls were there in the hall?

- (a) 200 (b) 100
(c) 400 (d) 150

56. A goldsmith used $\frac{2}{3}$ of his gold for ornaments, of that $\frac{4}{5}$ used for necklaces, and $\frac{3}{8}$ of those necklaces were defective. What fraction of his total gold was defective?

- (a) $\frac{1}{7}$ (b) $\frac{1}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

57. A weaver used $\frac{5}{6}$ of a roll of cloth to make sarees, with each saree required $5\frac{3}{4}$ m. If the roll contained $34\frac{1}{2}$ m of cloth, then how many sarees could be made?

- (a) 3 (b) 4 (c) 5 (d) 6

58. Calculate: $\left(\frac{7}{8} + \frac{14}{15}\right) \times \left(\frac{9}{10} + \frac{3}{5}\right) + \left(\frac{5}{6} \times \frac{2}{3}\right)$

- (a) $\frac{27}{16}$ (b) $\frac{81}{32}$
(c) $\frac{27}{32}$ (d) $\frac{81}{16}$

59. A quantity Q is multiplied successively by $\frac{4}{5}, \frac{3}{2}, \frac{5}{8}$ and then divided by $\frac{9}{10}$. The final result is 270. Find Q.

- (a) 324 (b) 225
(c) 250 (d) 200

60. A weaver cuts $\frac{2}{3}$ of a roll of cloth for shirts. From that she uses $\frac{3}{4}$ for coloured shirts. Then, she sells $\frac{3}{5}$ of those coloured shirts. If the roll contained 90 m of cloth, how many metres of coloured shirts remain unsold.

- (a) 12 m (b) 13 m
(c) 15 m (d) 18 m



Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1. Match the following :

List-I

List-II

- (P) Mayur studies for $5\frac{1}{3}$ hours daily, he study each subject for $\frac{4}{3}$ hours, how many subjects does he study? (1) 2
- (Q) The area of rectangular board is $10\frac{2}{3}$ sq. cm and length of board is $\frac{16}{3}$ cm then find its breadth (in cm). (2) 1
- (R) A piece of wire is $\frac{21}{3}$ m long. How many pieces of length $\frac{7}{3}$ m can be cut from the wire? (3) 4
- (S) What is the product of any fraction with its reciprocal? (4) 3
- (a) P-3, Q-2, R-1, S-4
(b) P-3, Q-1, R-4, S-2
(c) P-2, Q-1, R-4, S-3
(d) None of these

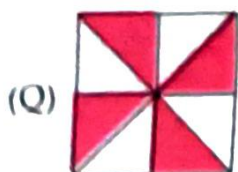
2. Match the shaded fraction of the figure given in List-I with the fraction in List-II.

List-I

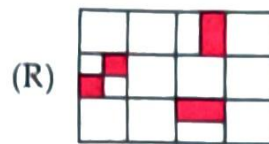
List-II



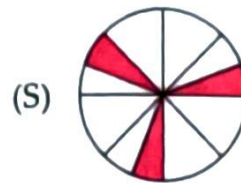
(1) $\frac{1}{2}$



(2) $\frac{1}{8}$



(3) $\frac{3}{16}$



(4) $\frac{2}{9}$

- (a) P-4, Q-2, R-3, S-1 (b) P-4, Q-1, R-3, S-2
(c) P-4, Q-1, R-2, S-3 (d) P-3, Q-1, R-4, S-2

3. Match the dividend-quotient relationship given in List-II for the fraction in List-I.

List-I

List-II

- (P) $\frac{585}{369} + \frac{465}{121}$ (1) Quotient > Dividend
- (Q) $\frac{612}{125} + \frac{81}{105}$ (2) Quotient = Dividend
- (R) $\frac{128}{512} + 1$ (3) Quotient < Dividend
- (a) P-3, Q-2, R-1 (b) P-3, Q-1, R-2
(c) P-2, Q-3, R-1 (d) P-1, Q-3, R-2



Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If assertion is false but reason is true.

1. **Assertion :** The product of two numbers is $15\frac{5}{6}$. If one of the numbers is $6\frac{1}{3}$, then the other number is $3\frac{1}{2}$.

Reason : $\left(\frac{a}{b} + \frac{c}{d}\right) = \left(\frac{a}{b} \times \frac{d}{c}\right)$, where $a, b, c, d \neq 0$ are whole numbers.

2. **Assertion** : If two fractions both are greater than 1, then their product is greater than both fractions.

Reason : When we multiply a number by a number greater than 1, the result becomes bigger.

3. **Assertion** : $\frac{1}{5}$ of $\frac{1}{6}$ of $\frac{2}{3}$ of 3600 is 90.

Reason : When we multiply two fractions, we are finding a part of a part.

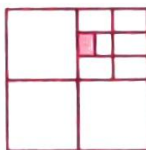
4. **Assertion** : When a number (the dividend) is divided by a proper fraction, the quotient will be greater than the dividend.

Reason : Dividing a number by the number less than 1 will always result in quotient smaller than dividend.

5. **Assertion** : The product of $\frac{1}{5} \times \frac{1}{8}$ is exactly same as the product of $\frac{1}{8} \times \frac{1}{5}$.

Reason : For any two fractions $\frac{a}{b}$ and $\frac{c}{d}$,
 $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$.

6. **Assertion** : The fraction of area of the whole square that the shaded region occupies is $\frac{1}{2}$ of $\frac{1}{6}$ of $\frac{1}{4}$.



Reason : When a square is divided into m equal parts then each smaller part represents $\frac{1}{m}$ of total area of the larger square.

7. **Assertion** : $\frac{3}{5} \times \frac{10}{9} = \frac{2}{3}$

Reason : While simplifying fractions, we cancel the common factors between numerator and denominator.

8. **Assertion** : When $\frac{4}{5}$ is multiplied by its reciprocal the product is $\frac{5}{5}$.

Reason : For a fraction $\frac{a}{b}$, $\frac{a}{b} \times \frac{b}{a} = 1$



Comprehension Type

PASSAGE-I : Brahmagupta gave rules for multiplying and dividing fractions as given below:

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}, \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

- The value of $\frac{2}{3} \times \frac{3}{5}$ equals to
 (a) $\frac{6}{15}$ (b) $\frac{2}{5}$
 (c) Both (a) and (b) (d) Neither (a) nor (b)
- The product of $\frac{a}{b}$ with reciprocal of $\frac{c}{d}$ is equal to
 (a) $\frac{a}{b} + \frac{c}{d}$ (b) $\frac{a}{b} \times \frac{c}{d}$
 (c) $\frac{c}{d} + \frac{b}{a}$ (d) $\frac{a}{b} \div \frac{c}{d}$
- For any fraction $\frac{a}{b} + 1$ equals to
 (a) $\frac{a}{b} \times 1$ (b) $\frac{b}{a}$
 (c) $\frac{a+1}{b}$ (d) None of these

PASSAGE-II : Three bottles of mango juice to be pour into glasses equally. A bottle contain

$1\frac{1}{2}$ litres and capacity of a glass is $\frac{3}{4}$ litre.

- What is the total quantity of juice in all 3 bottles?
 (a) $3\frac{1}{2}$ litres (b) $\frac{3}{2}$ litres
 (c) $\frac{9}{2}$ litres (d) $\frac{3}{4}$ litre

2. How many glasses can be fill with 3 bottles?
 (a) 6 (b) 7
 (c) 8 (d) 9
3. If one person drinks $\frac{1}{2}$ glass, how many persons can share the total glasses formed?
 (a) 12 (b) 14
 (c) 16 (d) 18

PASSAGE-III : If we take $\frac{a}{b}$ part of the whole

and then $\frac{c}{d}$ of the part is $\frac{c}{d}$ of $\frac{a}{b} = \frac{c}{d} \times \frac{a}{b} = \frac{c \times a}{d \times b}$.

1. Find $\frac{7}{8}$ of a day.
 (a) 20 hours (b) 21 hours
 (c) $\frac{7}{8}$ hours (d) 1 hour
2. Sapna earns ₹12000 per month. She spends $\frac{7}{8}$ of her income and deposits rest of the money in a bank. How much money does she deposit in the bank each month?
 (a) ₹10500 (b) ₹11000
 (c) ₹1500 (d) ₹5000
3. To buy a movie ticket, a person need to spend $\frac{5}{7}$ of the total money that he had. If he had ₹2100, then find the price of ticket.
 (a) ₹700 (b) ₹1500 (c) ₹1200 (d) ₹900

Subjective Problems

Very Short Answer Type

1. Simplify : $\frac{14}{25} \times \frac{35}{51} \times \frac{34}{49}$
2. Find the reciprocal of:
 (i) 5 (ii) $\frac{1}{3}$
 (iii) $\frac{7}{9}$ (iv) $4\frac{1}{5}$
3. Find $2\frac{1}{5} + 1\frac{1}{5}$.

4. Divide $\left(\frac{1}{2} \text{ of } \frac{4}{3}\right)$ by $\frac{1}{3}$.
5. Divide the sum of $3\frac{1}{4}$ and 3 by $2\frac{3}{5}$.
6. When one number in a multiplication is between 0 and 1, how does the product compare to the other number?
7. If 1 copper pana = 30 cowrie shells, then find the fraction that represents 1 cowrie shell in terms of copper pana.
8. A book consists of 216 pages. During last week Vikas read $\frac{3}{4}$ of the book. How many pages did he read?
9. Aarav plants $\frac{1}{2}$ of his garden with roses and $\frac{2}{3}$ of that with red roses. What part of the garden has red roses?
10. Megha's salary is $\frac{3}{7}$ of Mahesh's salary. If Megha earns ₹12600. What is Mahesh's salary?

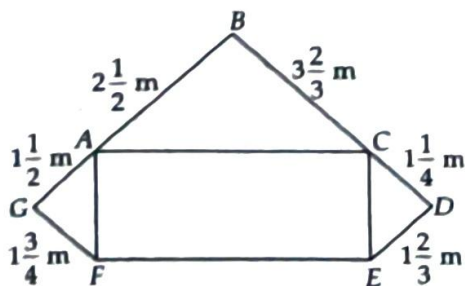
Short Answer Type

1. Find $\frac{1}{4}$ of
 (a) $\frac{1}{4}$ (b) $\frac{3}{5}$ (c) $\frac{4}{3}$
2. Simplify : $2\frac{2}{17} \times 7\frac{2}{9} \times 1\frac{33}{52}$
3. Multiply the following and reduce to lowest form and convert into a mixed fraction:
 (i) $7 \times \frac{3}{5}$ (ii) $4 \times \frac{1}{3}$
4. Observe the given pattern and write a general statement for the result.
 $\left(1 + \frac{1}{2}\right)$
 $\left(1 + \frac{1}{2} + \frac{1}{3}\right)$
 $\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right)$

- Divide 36 by $6\frac{2}{3}$ and subtract the quotient from $7\frac{3}{5}$.
- A tin contains 18 kg ghee. After consuming $\frac{2}{3}$ of it, how much ghee is left in the tin?
- A goldsmith used $\frac{5}{8}$ of a gold bar to make bangles and $\frac{1}{4}$ of the remaining bar to make earrings. What fraction of bar remains?
- A coin is made from $\frac{1}{4}$ of silver, $\frac{1}{4}$ of gold and $\frac{1}{2}$ of copper. If 2 gram copper is used to make the coin, then find the total weight of coin.
- A clay artist uses $\frac{1}{2}$ kg of clay for each model. If she has $2\frac{1}{2}$ kg, how many models can she make?
- Reenu spends $\frac{4}{5}$ of her income on household expenses. Her monthly income is ₹15000. How much does she save every month?

Long Answer Type

- The perimeter of the given figure is 18 m and the area of rectangular part ACEF is $8\frac{1}{2}$ sq. m. Find the length and breadth of the rectangular part.



- Find the reciprocal of the sum of $1\frac{7}{9}$ and $1\frac{9}{7}$.

- A mat is divided into 54 colourful blocks. If $\frac{5}{18}$ of the mat is red in colour, $\frac{1}{3}$ of the mat is blue and remaining is yellow, find the fraction of yellow blocks and number of blocks of each colour.

- Some pictures (a) to (c) are given. Tell which of them show:

(i) $3 \times \frac{1}{5} = \frac{3}{5}$ (ii) $2 \times \frac{1}{3} = \frac{2}{3}$

(iii) $3 \times \frac{3}{4} = 2\frac{1}{4}$



- Reena has $4\frac{1}{2}$ apples. She gives $1\frac{3}{4}$ apples to her friend and cuts the rest into $\frac{1}{4}$ apple pieces to serve.

- How many apples does she have left after giving $1\frac{3}{4}$ to her friend?
- How many $\frac{1}{4}$ apple pieces can she make from the remaining apples?



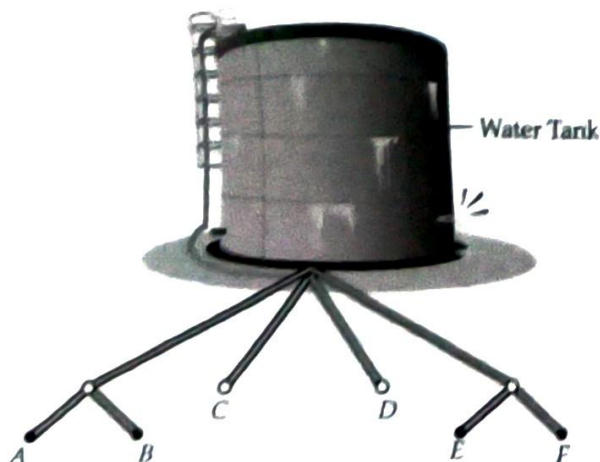
Integer/Numerical Value Type

- How many pieces of equal size can be cut from a rope of 30 metres long, each measuring $3\frac{3}{4}$ metres?
- By what number should $2\frac{3}{5}$ be multiplied to get $5\frac{1}{5}$?

- A fraction is in its lowest form if its numerator and denominator have no common factor other than ____.
- $\frac{2}{8}$ of a day = ____ hours.
- If the cost of $4\frac{1}{4}$ kg of sugar is ₹ 68, then find the cost of 1 kg of sugar (in ₹).
- What should be the sum of numerator and denominator if $\frac{5}{8}$ is divided by $\frac{25}{32}$?
- The value of $\frac{1}{5}$ of $\frac{2}{3}$ of $\frac{1}{6}$ of 2250 is ____.
- A farmer divides 12 kg of seeds equally into bags of $\frac{2}{3}$ kg each. How many bags will he fill?
- Find the product of reciprocal of $\frac{2}{5}$ with reciprocal of $\frac{5}{2}$.
- If a matchbox costs ₹ $1\frac{1}{4}$. Then, how many matchboxes can be bought in ₹ 10?

Case Based Questions

Case I : A water tank supplies water to several villages. At first junction, the main pipe splits into 4 equal branches. The left most and the right most again splits into 2 equal branches at next junction. Finally, the tank supplies water to 6 villages A, B, C, D, E and F.



Based on the above information, answer the following questions.

- After first split, what fraction of water goes into each of the 4 main branches?
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$
- What fraction of total supply reaches to village B?
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$
- The village C used $\frac{2}{3}$ of the water received, for irrigation. So, what fraction of total supply is used for irrigation in village C?
(a) $\frac{2}{9}$ (b) $\frac{3}{7}$ (c) $\frac{1}{6}$ (d) $\frac{1}{8}$
- If the capacity of water tank is 8000 litres, then the quantity of water supplies to village F is
(a) 1000 litres (b) 2000 litres
(c) 2500 litres (d) 3800 litres
- Which of the given village received maximum quantity of water?
(a) A (b) B (c) C (d) E

Case II : Raghav's grandfather gave him an old coin collection. It had rupees, aanas and paise – coins used in India long ago. He told Raghav that 1 rupee = 100 paise and 4 aanas = 25 paise.

Raghav was curious and he start counting the coins.

Based on the above information, answer the following questions.

- 1 paisa is equal to what fraction of a rupee?
(a) $\frac{1}{50}$ (b) $\frac{2}{75}$ (c) $\frac{1}{75}$ (d) $\frac{1}{100}$
- 1 paisa is equal to what fraction of an aana?
(a) $\frac{4}{25}$ (b) $\frac{2}{25}$ (c) $\frac{3}{50}$ (d) $\frac{1}{16}$
- How many aanas is equal to 1 rupee?
(a) 8 (b) 10 (c) 12 (d) 16

4. 1 aana is equal to what fraction of a rupee?

- (a) $\frac{1}{12}$ (b) $\frac{1}{16}$
(c) $\frac{1}{10}$ (d) $\frac{1}{8}$

5. If Raghav has 3 coins of 8 aanas, how many rupees does he have?

- (a) $\frac{5}{2}$ (b) $\frac{3}{4}$ (c) $1\frac{1}{2}$ (d) $1\frac{1}{4}$

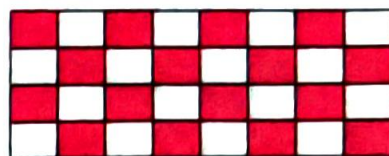
Case III : A baker has $8\frac{1}{2}$ kg cake. He cuts pieces of $\frac{1}{8}$ kg each. He sells $\frac{3}{4}$ of the pieces in the morning. The cost of 1 piece of cake was ₹ 20. The remaining pieces sold by the baker in the evening for ₹ 15 each.

Based on the above information, answer the following questions.

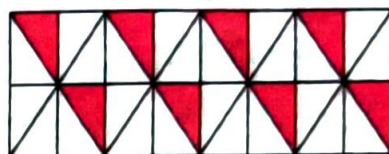
- How many pieces are cut from the total cake?
(a) 51 (b) 68 (c) 65 (d) 52
- How many pieces were sold in the morning?
(a) 51 (b) 41 (c) 31 (d) 21
- What fraction of total pieces were sold in the evening?
(a) $\frac{2}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$
- Amount received from morning sale is
(a) ₹ 845 (b) ₹ 2010
(c) ₹ 1200 (d) ₹ 1020
- Amount received from evening sale is what fraction of total amount received from the sale?
(a) $\frac{1}{5}$ (b) $\frac{1}{4}$ (c) $\frac{1}{3}$ (d) $\frac{1}{6}$

Case IV : In Art class, three students Maya, Mohit and Raja were working on a large rectangular shape canvas. The teacher asked them to divide the whole canvas into 8 equal parts and further divide the each part into 4 more equal parts and shade few of them to make a pattern on the canvas.

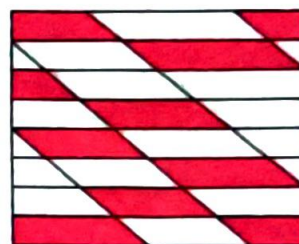
The following is the pattern made by them.



Maya



Mohit



Raja

Based on the above information, answer the following questions.

- Who among the three does not follow the instructions?
(a) Maya (b) Mohit
(c) Raja (d) None of these
- What fraction of Maya's canvas represents the each smaller part?
(a) $\frac{1}{16}$ (b) $\frac{1}{8}$
(c) $\frac{1}{28}$ (d) $\frac{1}{32}$
- What fraction of pattern made by Maya is shaded?
(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{6}$
- What fraction of canvas is shaded in the pattern made by Mohit?
(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$
- If minimum shaded fraction should be $\frac{1}{4}$, how many triangles does Mohit need to shade more?
(a) 3 (b) 2 (c) 1 (d) 0

Case V : Ramesh ordered a pizza from online food site. The pizza have 8 equal parts (slice). Ramesh ate $\frac{3}{8}$ part of pizza. His friend Ashish ate $\frac{4}{8}$ part of pizza and remaining pizza was eaten by his sister Sita.



Based on the above information, answer the following questions.

1. If the cost of pizza is ₹ 80, then Sita share is

(a) ₹ 20 (b) ₹ 10 (c) ₹ 15 (d) ₹ 8

2. Who ate the most of the part of pizza?

(a) Ramesh (b) Sita
(c) Ashish (d) None of thee

3. If the cost of pizza is ₹ 80, then Ramesh share is

(a) ₹ 30 (b) ₹ 15 (c) ₹ 60 (d) ₹ 90

4. The pizza eaten by Sita is shown by which of the following options?

(a)  (b)  (c)  (d) 

5. Which of the following is equivalent to Ashish part of pizza?

(a) $\frac{2}{4}$ (b) $\frac{1}{2}$

(c) Both (a) and (b) (d) None of these